

Ali Sao

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EDUCATION

University of California, Irvine

Expected Jun. 2027

B.S. in Computer Engineering

Irvine, CA

Relevant Coursework: Data Structures & Algorithms, System Software, Advanced Databases, Operating Systems, Computer Architecture, Embedded Systems, Digital Logic Design

EXPERIENCE

Software Engineering Intern (Software Integration)

Jun. 2026 – Present

Panasonic Avionics Corporation

Irvine, CA

- Developed **Bash automation scripts** on **Linux systems** to process and export aircraft **fault** and diagnostic data for airline maintenance teams.
- Improved fault-report retrieval performance by implementing **caching** and **async data pipelines** in **Node.js** services, reducing large diagnostic export times by **40%**.
- Developed automated **testing**, **CI/CD**, and deployment workflows using **Selenium**, **Jest**, **Bash**, and internal Linux tooling, improving reliability of maintenance software releases.
- Monitored and analyzed aircraft **system logs** and integrated backend **fault APIs** with **SQL**-based data services to support diagnostic workflows and software delivery for airline operators.

Full-Stack Software Engineering Intern

Sep. 2025 – Jan. 2026

Tegore.ai (YC X25)

San Francisco, CA

- Developed a real-time **AI streaming pipeline** using **Python**, **FastAPI**, **React**, and **WebSockets**, reducing first-token response latency to under **250ms** for live tutoring sessions.
- Built **Docker**-based backend infrastructure with **PostgreSQL**, **FastAPI**, and **Kubernetes** deployment environments to support scalable AI tutoring and multi-user API workloads.
- Built early **RAG pipelines** and AI lesson-generation systems to ingest textbook content, process educational data, and automate personalized tutoring workflows.
- Developed backend APIs and deployment workflows for AI-powered services using **Python**, **Node.js**, and containerized Linux environments.

Undergraduate Researcher

Sep. 2024 – Present

Wayne Hayes Lab, University of California Irvine

Irvine, CA

- Scaled a neural-network alignment platform into a public research web application by building distributed backend infrastructure with **Node.js**, **Redis**, and **BullMQ**, supporting 1,000+ research users.
- Automated long-running computational biology workloads by developing **Python** and JavaScript services for **job scheduling**, retries, persistent queues, and distributed task processing.
- Deployed and maintained **Linux-based virtual machine infrastructure** using **Apache** and backend web services to host and support large-scale computational research workloads.

PROJECTS

Senior Design Project – High-Frequency Trading Visualization System

Sep. 2025 – Present

Project — React, Node.js, Verilog, C

Irvine, CA

- Built a high-performance **React dashboard** to visualize real-time trading signals and market state, using reusable visualization components optimized for streaming data.
- Built a low-latency data pipeline connecting **Verilog** and **C**, streaming FPGA-generated market signals over **UART** into a **C-based order book engine**.
- Developed a **WebSocket streaming interface** in **C** to transmit order book state and trading metrics to a **React UI** while maintaining responsive real-time visualization.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Bash, SQL, C/C++, HTML, CSS

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Express, Tailwind CSS, BullMQ, WebSockets, Firebase

Infrastructure & DevOps: Linux, Docker, Kubernetes, PostgreSQL, Redis, Apache, NGINX, CI/CD, GitLab, Jenkins, Prometheus, Google Cloud Platform

Developer Tools: Git, Selenium, Jest, VS Code